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Advances in Space Research xxx (2020) xxx–xxx

ADVANCES IN SPACE RESEARCH (a COSPAR publication)

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Seismo ionospheric anomalies in Turkey associated with $M_w \geq 6.0$ earthquakes detected by GPS stations and GIM TEC

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Received 23 August 2019; received in revised form 26 February 2020; accepted 6 March 2020

Abstract

Earthquake (EQ) anomalies in the form of enhancement and depletion in ionospheric Total Electron Content (TEC) from Global Positioning System (GPS) may considerably alarm about short and long term precursors of the impending main shock. In this paper, TEC anomalies are investigated from permanent GPS ground-stations in Turkey associated to $M_w \geq 6.0$ EQs occurred in 2011–2012. Temporal and spatial analyses of TEC at 2 h sampling have shown significant evidences about EQ induced ionospheric anomalies during 10–14 h of UT (Universal Time) within 5 days before M_w 6.0 Greece, and M_w 7.1, Turkish EQ. Spatial analyses have manifested arrival of TEC anomalies at UT = 10 h to epicenter of both EQs, which linger above epicenter during UT = 12–14 h and left seismogenic zone after UT = 14 h before every EQ during $K_p < 3$ and $Dst = 0$ nT. Meanwhile, a geomagnetic storm ($Dst < -100$ nT) induce perturbation two days after the M_w 7.1 Turkish EQ, showing no relation with epicenter during spatial analysis. It also shows that TEC can be useful to distinguish geomagnetic storm variations to successfully detect EQ precursors. These anomalies during quiet storm ($K_p < 3$; $Dst = 0$ nT) conditions may be effective to link the lithosphere and ionosphere in severe seismic zones to detect EQ precursors before future EQs. Interpretation of TEC anomalies and it enhancements over EQ epicenters during UT = 12–14 h for both EQs have shown that EQs anomalies only occurred in particular time. Whereas, geomagnetic storm effect occurred during whole abnormal day over the Earth.

Keywords: GPS TEC; Earthquakes; Geomagnetic storm

1. Introduction

The phenomena around epicenter that produces ionospheric anomalies has been widely reported before and after large EQs from the analysis of GPS TEC above seismically active regions on Earth and it provides significant insight on the underlying mechanism of lithosphere-ionosphere coupling (Fujiwara et al., 2004; Liu et al., 2006; Shah and Jin, 2015; Daneshvar and Freund, 2017; Tariq et al., 2019). However, phenomena that precede large magnitude EQs essentially above active seismogenic zone have been reported more widely with available GPS TEC measurements than post seismic ionospheric anomalies (Liu et al., 2004; Shah et al., 2019a,b; Ahmed et al., 2018). A large effort has been reported to detect short and long term seismic anomalies before impending main

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https://doi.org/10.1016/j.asr.2020.03.005

0273-1177/

Please cite this article as: M. Shah, S. Inyurt, M. Ehsan et al., Seismo ionospheric anomalies in Turkey associated with $M_w \geq 6.0$ earthquakes detected by GPS stations and GIM TEC, Advances in Space Research, https://doi.org/10.1016/j.asr.2020.03.005

shock from ionospheric and atmospheric indices, including GPS TEC and other remote sensing measures (Píša et al., 2011; Shah et al., 2019b). All these variations in ionosphere and atmosphere were presented as bona fide pre EQ precursors within seismogenic period. On the other hand, there are also many reports against the existence of seismic anomalies (e.g., Geller et al., 1997; Rishbeth, 2007); some even strongly stated that EQ prediction is impossible with TEC (Thomas et al., 2017).

Beside different ionosphere anomalies, synchronized and collocated EQ induced TEC anomalies have been reported by different statistical and mathematical procedures (Dautermann et al., 2007; Afraimovich and Astafyeva, 2008). For example, seismic induced ionospheric anomalies are distinguished from that originated by geomagnetic storm at different locations on the Earth during 1998-2014 Shah and Jin (2015). Furthermore, they reported that GPS TEC anomalies depended on magnitude and hypocentral depth of future EQ and can also be attributed to different fault system. Similarly, Liu et al. (2001) observed significant GPS TEC within 10 days before the Chi-Chi EQ and found significant electron clouds associated with this EQ in spatial TEC maps during seismic preparation period. For more understanding of different morphological characteristics of EQ related anomalies, many analyses of ionospheric anomalies is performed on regional and local scale (Pulinets and Legen'ka, 2003; Saroso et al., 2008; Kon et al., 2011; Le et al., 2011; Yao et al., 2012; Ke et al., 2016).

EQ is a mechanical process that occur due to tectonic forces inside the Earth during EQ preparation period. Due to these tectonic stresses, Earth crust is under high level of stress that effectively cause catastrophic rock rupture (Freund et al., 2007; 2009). According to Daneshvar and Freund (2017), the association of ionosphere anomalies prior to any EQ may be based on the mechanical stress inside the rock column which further originate perturbations on Earth surface, and subsequently propagated to atmosphere over epicenter. The other explanation for triggering ionospheric variations above epicenter is due to enormous emanation of Radon gas from seismogenic zone and raising to ionosphere by the mechanism of lithosphere atmosphere ionosphere coupling within EQ preparation period (Pulinets et al., 2006; Pulinets and Ouzounov, 2011).

We analyzed ionospheric anomalies in temporal series from GPS TEC associated with two $M_w \geq 6.0$ EQs during 2011–12 from permanent GPS ground-stations in Turkey and verified by spatial GIM series. We implement different statistical procedures to distinguish seismo ionospheric anomalies specifically triggered by the impending main shock. The ionospheric enhancement related to geomagnetic storms is discussed in detail with enough evidences. Section 2 introduces the data and method, while Section 3 the results. In Section 4, we correlate the ionosphere anomalies with impending EQs, and provide sufficient evidences from previous reports. Finally, in Section 5, we provide our conclusions.

2. Data and method

In this paper, we study GPS TEC anomalies related to two $M_w \geq 6.0$ EQs during 2011–12 from permanent GPS ground-stations in Turkey. Information about EQs is retrieved from USGS via http://www.earthquake.usgs.gov/earthquakes/search and presented in Table 1. Similarly, geomagnetic storm, AE, D*st* and K_p indices are obtained from International Service of Geomagnetic Indices http://isgi.unistra.fr. The Auroral Electrojet (AE) index is introduced by Davis and Sugiura (1966), which is the measure of global electrojet activity in the auroral zone. Similarly, the D*st* index denotes the global magnetic activity by measuring the intensity of the Earth equatorial electrojet (Earth's ring currents). It is derived from a network of near-equatorial geomagnetic observatories on the Earth. The K-index also quantifies the Earth's magnetic field disturbances of the horizontal component (Bartels et al., 1939). It is calculated on the basis of 3-hour measurements from globally distributed magnetometers in the range of 0–9, where 1 being quiet storm activity and 5 or more shows intense geomagnetic storm (Loewe and Prölss, 1997). The main purpose of including geomagnetic storm in this study is to distinguish whether ionospheric anomalies are induced by storm or seismic process.

The EQ prone regions of all the EQs are estimated by the below formula;

$$R = 10^{0.43 M_w} \tag{1}$$

Where M_w is EQ magnitude and R is the preparation zone's radius (Dobrovolsky et al. (1979)). It can be seen from Equation (1) that the radius of sesimogenic zone is dependent on EQ magnitude. For example, strong EQ has large preparation zones and vice versa. For this study, all the stations operated within the EQ preparation zone are shown in Fig. 1.

EQs anomalies are investigated by spatial maps acquired from CODE TEC GIMs ftp://cddis.gsfc.nasa.gov/pub/gps/products/ionex. GIMs for all anomalous days are analyzed over epicenter to monitor the ionospheric perturbations. The TEC GIMs have temporal resolution of 2 h and spatially extension includes (2.5°) latitude, (5°) longitude (Feltens, 2003; Hernández-Pajares et al., 2009; Liu et al., 2016; Roma-Dollase et al., 2018). Moreover, Vertical TEC (VTEC) is retrieved from permanent GPS ground-stations in Turkey and investigated along with spatial data. VTEC observations from permanent GPS ground-stations in Turkey within seismogenic zones are examined to detect the consequent relationship of impending EQ and ionosphere. For each event, three available GPS stations are selected within Dobrovolsky et al. (1979) region. In this scheme, TVAN, FETH, FINI, MUGL, MURA, OZAL stations are selected.

The Slant TEC (STEC) retrieved from GPS stations occur as number of electrons within (1 × 1) square meter tube. It is calculated along the ray path of the observed

Table 1
EQ information and associated pre/post TEC precursors.

	EQ	Date and time (UT)	Mw	Depth (km)	Lat (°)	Long (°)	Pre EQ day	Post EQ	GNSS Stations Information Latitude and Longitude (degree)		
1	Dodecanese Islands, Greece	10-06-2012/ 12:44:16	6.0	35.0	36.42	28.880	EQ day	–	FETH (36.6262 29.1237)	FINI (36.3022 30.1464)	MUGL (37.2163 28.3644)
2	Eastern Turkey	23-10-2011/ 10:41:23	7.1	18.0	38.72	43.508	–2	+2	MURA (38.9901 43.7631)	OZAL (38.6573 43.9886)	TVAN (38.5294 42.2904)

signal in the unit of TEC, where 1 TECU = 10^{16} electron/m^2 (Li et al., 2012), by the following equation:

$$\mathrm{STEC} = \frac{f_1^2 f_2^2}{40.28(f_1^2 - f_2^2)}(L_1 - L_2 + \lambda_1(N_1 + b_1) - \lambda_2(N_2 + b_2) + \epsilon) \tag{2}$$

$$\mathrm{STEC} = \frac{f_1^2 f_2^2}{40.28(f_1^2 - f_2^2)}(P_1 - P_2 - (d_1 - d_2) + \epsilon) \tag{3}$$

where STEC, f_1 and f_2 are Slant Total Electron Content, carrier phase frequency at one end (1 × 1) square meter tube of GPS signals and frequency at second end, respectively. The pseudo-range delay of the signal is denoted by P_1 and P_2 and L carrier phase observation of delay path. The wavelength of GPS signal is λ. N is ambiguity of signal and b is the instrumental biases. d_1 and d_2 are the differential code biases and ϵ denotes random residuals of ray path. Furthermore, STEC is transformed into VTEC by the following equation (Klobuchar, 1987);

$$\mathrm{VTEC} = \mathrm{STEC} \times \cos\left(\mathrm{arcsine}\left(\frac{R\sin Z}{R + H}\right)\right) \tag{4}$$

In this equation, R denotes the Earth radius and H is the elevation of ionosphere's upper limit. Similarly, Z is the satellite elevation angle (Heki and Enomoto, 2013).

In order to obtain VTEC from the GPS observations, RINEX data of each station is processed using STEC calibration method (Ciraolo et al., 2007). The data was obtained from website https://www.tusaga-aktif.gov.tr/Sayfalar/SistemeGiris.aspx. In processing stage, cut-off angle is taken 10° and VTEC values obtained at two-hour temporal resolution.

In order to distinguish anomalous signals in VTEC measurements, a descriptive statistical analysis is adopted on daily bi-hourly TEC; the one implement in Liu et al. (2001). The median ($\bar{x}$) of 10 days before and after the observed day is calculated to construct bounds. Furthermore, to quantify the anomalous variation, first (lower) and third (upper) quartiles are also calculated, which are denoted by LQ and UQ. As compared to VTEC normal distribution with mean (μ) and standard deviation (σ), these bounds are expected to be μ and 1.34σ, respectively (Klotz and Johnson, 1983). In this paper, we estimated the lower bound by $\bar{x} - \mathrm{LQ}$ and similarly, upper bound by $\bar{x} + \mathrm{UQ}$. After this, if the original VTEC occur outside of either bound, the variations are anomalous with 65–70% confidence that a positive (moved up beyond upper bound) or negative (moved down below lower bound) signal is detected (Liu et al., 2001).

The difference of TEC from median is calculated for all stations for every EQ by the following equation

$$\Delta\mathrm{VTEC}_{max} = (\mathrm{VTEC}_{current} - \mathrm{VTEC}_{median})/\mathrm{VTEC}_{median} \tag{5}$$

The difference of every station within seismogenic zone of the EQ shows clear variations on abnormal day. The abnormal variations in temporal TEC is further check in spatial maps. Then, variations on anomalous day in every UT hour are correlated with impending EQ. Some UT hours during temporal analysis overlap beyond the confidence bounds in the analysis of GNSS TEC around the epicenters as precursor. Spatial TEC maps also show abnormal TEC clouds over the epicenters in the same UT hours as temporal analysis.

3. Results and description

In this paper, ionospheric anomalies detected by GPS ground-stations in Turkey are being studied for correlation with two $M_w \geq 6.0$ EQs during 2011-12. Fig. 1 shows the geographical location of EQs epicenter and GPS station and EQs seismogenic zones. In addition, we examine TEC variations due to geomagnetic storm in all events and no explicit relationship of triggering ionospheric variation has been found. The pre and post seismic ionospheric enhancement is summarized in Table 1.

The ionospheric anomaly is studied at 2 h sampling interval of TEC for three stations on EQ day associated to the M_w 6.0, Greece EQ. We investigate GPS TEC observables in the Dobrovolsky et al. (1979) region (Fig. 3). Diurnal variation manifests abnormal TEC value during daytime beyond the upper confidence bound within seismogenic zone. This shows a positive day time TEC perturbation mainly associated to main shock. Geomagnetic storm indices show no association of abnormal variation, as they show quiet storm on EQ day. The temporal enhancement is further checked by a series of GIM's over the epicenter on June 10, 2012 associated with M_w 6.0 EQ (main shock day). The bi-hourly spatial maps show commencement of TEC enhancement at UT = 12 h, which

Fig. 1. Geographical location of two $M_w \geq 6.0$ EQs in Turkey with Dobrovolsky region in red circles. The epicenters are denoted by red stars and GPS ground-stations in Turkey are represented by black asterisk. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

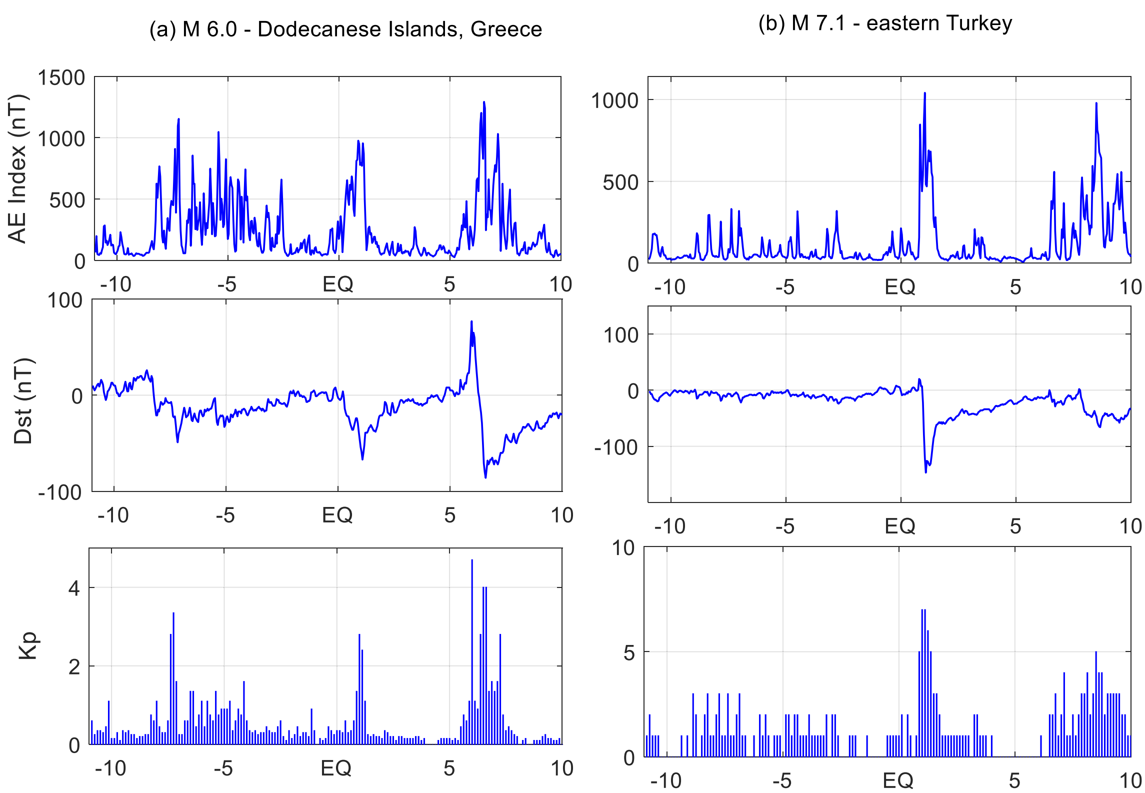

Fig. 2. Geomagnetic storm conditions before and after two $M_w \geq 6.0$ EQs from AE, D*st* and K_p indices. The EQs information is given on top of all panels.

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linger over epicenter during UT = 12–14 h and last until UT = 14 h (Fig. 4).

We find positive anomaly associated with M_w 6.0 EQ, which occur during $K_p < 3$ and $Dst = 0$nT. The positive anomaly will be the VTEC value beyond the upper bound, as per Liu et al. (2004) definition. Similarly, they showed negative anomaly as low VTEC value from lower confidence bound in Taiwan region. The anomaly in our study is likely due to the main shock, as geomagnetic storm is quiet represented by $K_p < 3$ and $Dst = 0$ nT. Similarly, original TEC show no enhancement in response to moderate geomagnetic storm at 7 days before M_w 6.0 EQ. Temporal TEC follows normal diurnal pattern within confidence bounds on this moderate geomagnetic storm day. This shows that every geomagnetic storm cannot cause variations in TEC. Burešova and Laštovička (2007) reported about 20–25% ionosphere anomalies with strong pre geomagnetic storm enhancements and observed only 20–25% pre storm enhancements. It means that chance of pre geomagnetic storm enhancement with moderate storm is very low. This shows that pre EQ enhancement on June 10, 2012 (EQ day before main shock) is mainly due to impending EQ.

Association of ionospheric anomalies with EQ are also studied in a similar way for the M_w 7.1 in Turkey. The variations in VTEC within Dobrovolsky et al. (1979) region show ionospheric anomaly 2 days before main shock (Fig. 5). Fig. 5 also show significant perturbation 2 days after the main shock day, which is attributed to geomagnetic storm, as shown by Dst Index. During the main phase Dst is less than -100nT and $K_p > 5$ (Fig. 2). This storm causes variations of >5 TECU and its impact on ionosphere is bigger than that given by EQ. The EQ causes only variation of ≤ 5 TECU. However, this variation may be more for large magnitude EQ. Here, we study only M_w 7.1 EQ to differentiate TEC anomaly. It is possible that large magnitude EQ may cause more variations in TEC as compare to low magnitude EQ.

Ionospheric enhancement before and after occurrence of main shock is further examined in GIM TEC maps over epicenter (Figs. 6 and 7). Dense clouds of enhanced TEC trespassed during UT = 10–14 h on October 21, 2011 (2 days) before the main shock of M_w 7.1, Turkey's EQ (Fig. 6). We attribute these anomalous clouds of TEC to future EQ, as it enhancement is more over epicenter than in the characteristic two-crest equatorial region. Geomagnetic and solar activity shows low during this period (Fig. 2). On the other hand, spatial TEC maps after EQ on October 25, 2011 (2 days after) showed no close relationship with epicenter and followed normal equatorial drift path (Fig. 7). This shows that TEC anomaly on October 25, 2011 is not triggered by EQ and most probably due to geomagnetic storm.

The analysis of VTEC during 10 days before and after for two $M_w \geq 6.0$ EQs show rapid enhancement as detected in bounds method on suspected days (Fig. 8). The abnormal days are more prominent from the other days. The positive anomalies monitored before and after each EQ

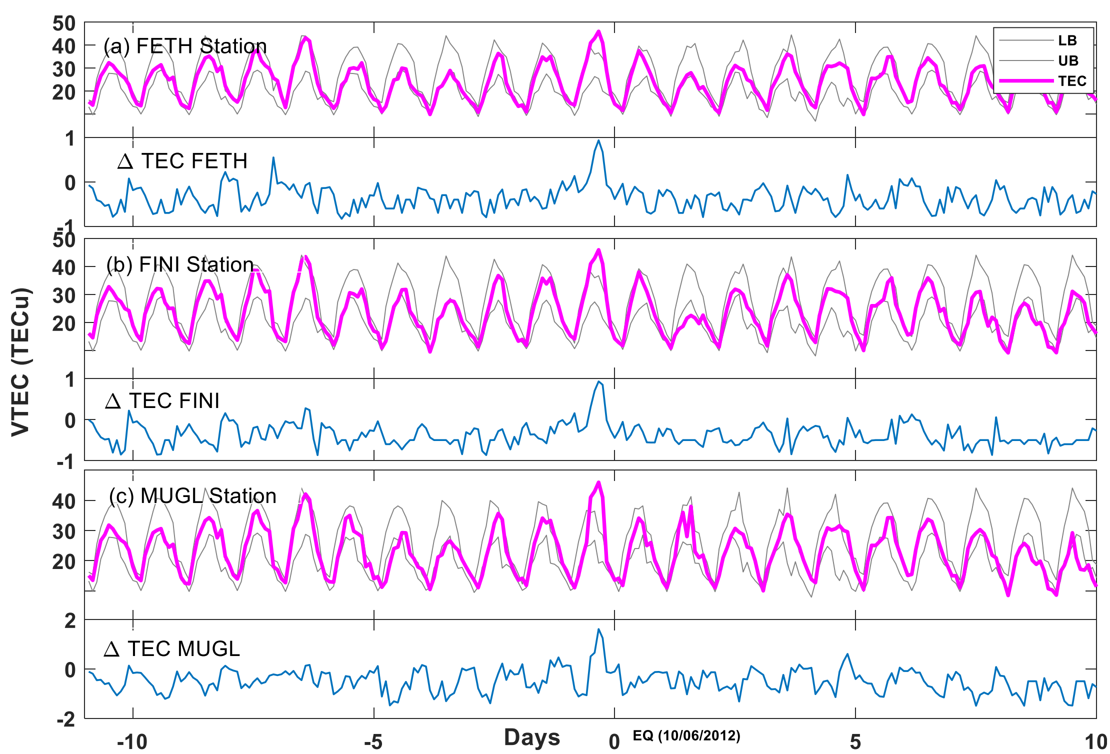

Fig. 3. Temporal VTEC from three GPS ground-stations in Turkey for analysis of anomalous values before and after M_w 6.0 Greece EQ (see Table 1 for EQ details). The original VTEC and confidence bounds are showed by pink and grey colors, respectively. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article)

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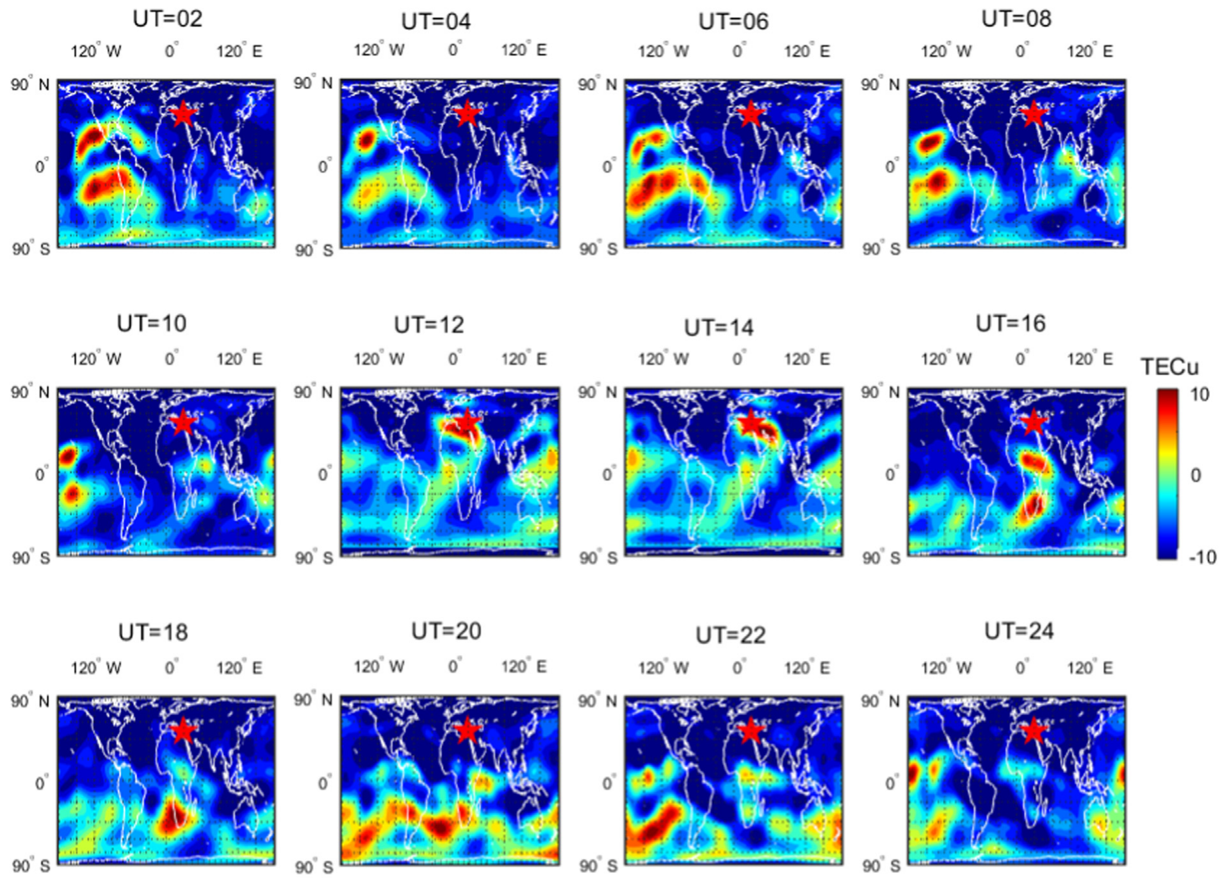

Fig. 4. Spatial GIM TEC maps on global scale for M_w 6.0 Greece EQ on June 10, 2012 (main shock day) in the form of differential values for observed day minus mean of one month. The red filled star is for annotation of epicenter, whereas UT is shown as title on each panel. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

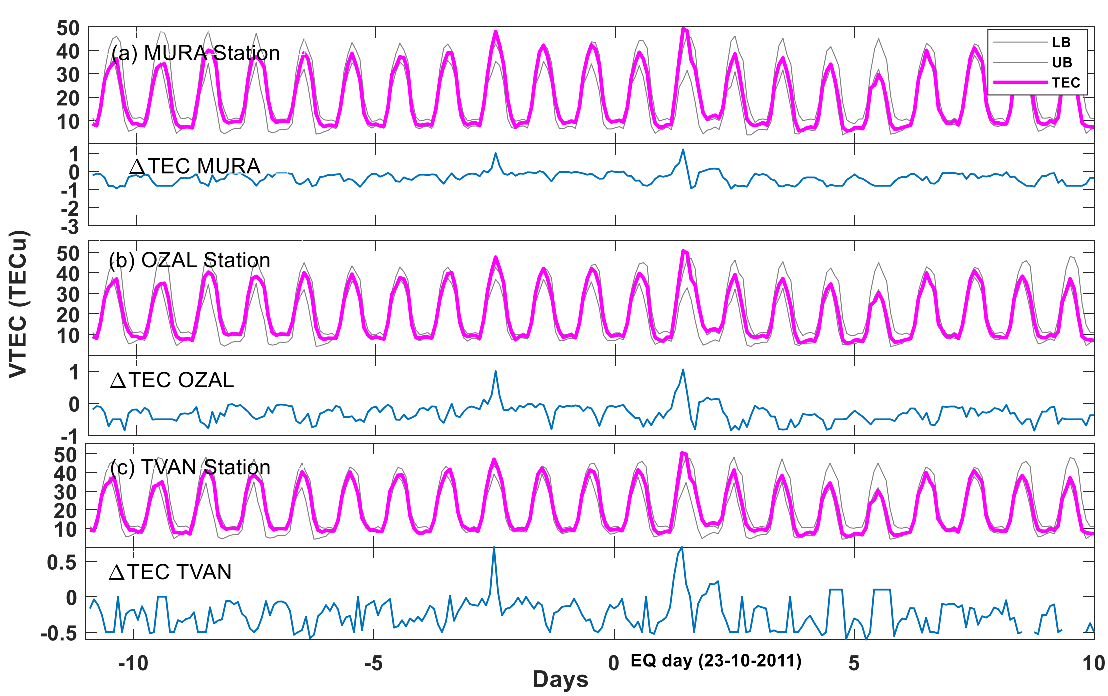

Fig. 5. Temporal VTEC and confidence bounds of median and associated IQR from three GPS ground-stations in Turkey for EQ occurred on October 23, 2011, where TEC values are acquired from three stations (name of stations mention in each panel).

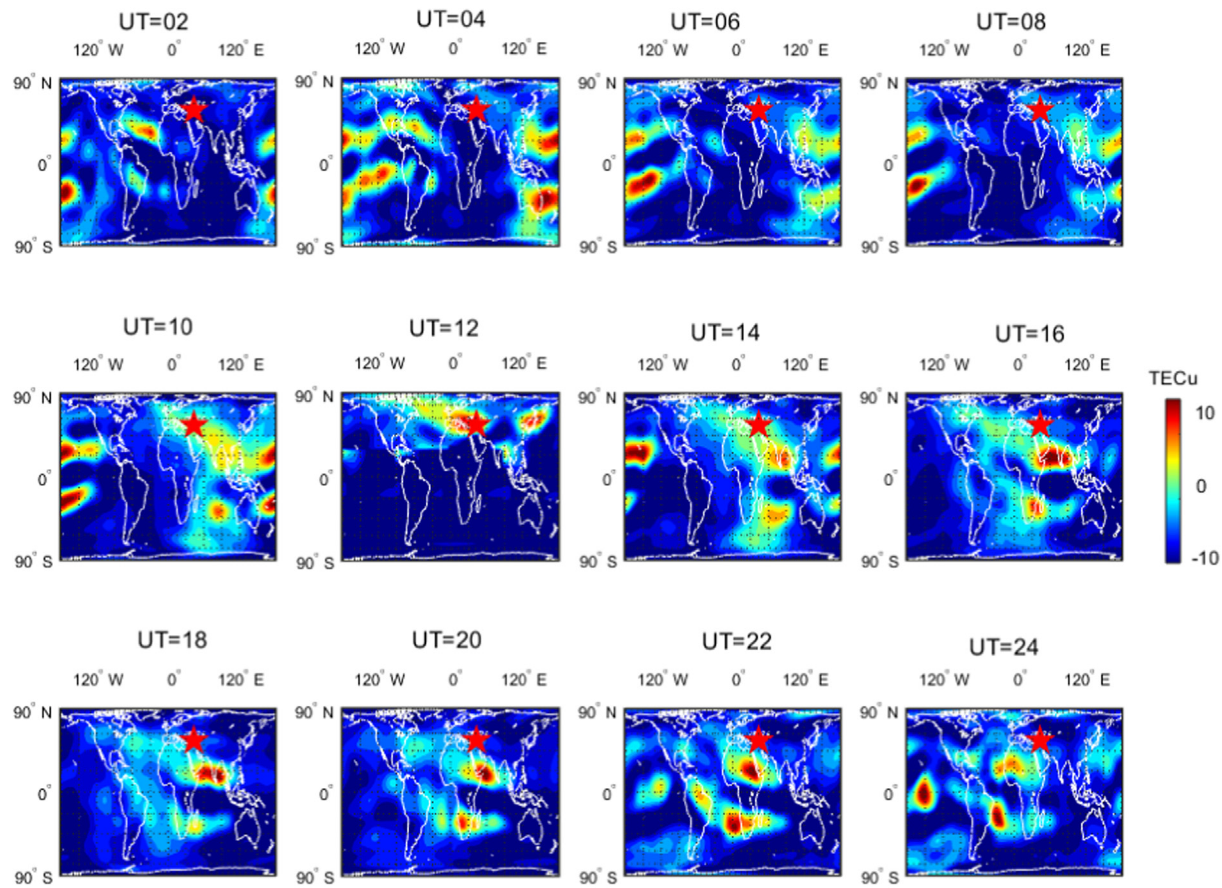

Fig. 6. Spatial TEC before Turkish EQ on October 21, 2011 (2 days before main shock).

in individual observation are also recorded and it provide evidence to justify our statistical bound manifestation. Positive anomalies are recorded on 0 (main shock day) and 2 days before EQs in Greece and Turkey, respectively (Fig. 9). TEC during UT = 12–14 h showed enhancements on suspected days (Fig. 9). On the other hand, TEC during UT = 16 h showed no profound enhancement before and after M_w 6.0 (Greece EQ) main shock. A little variation trigger by M_w 7.1, Turkish EQ report but not evident as UT = 12–14 h. This shows that EQ associated variations may or may not occur but It should be further investigated and described as previously in spatial TEC measurement over EQ epicenters.

4. Discussion

This paper shows the analysis of TEC for enhancement related to two EQs aim to study the coupling between lithosphere and ionosphere. We found significant perturbations in the study of TEC from confidence bounds from three different GPS ground-stations in Turkey in the case of each EQ. The positive TEC anomalies associated with the main shock in all case studies provide verification of EQ related ionospheric anomalies (Figs. 3 and 5). We show EQ anomalies within 5 days before two large magnitude EQs. Similarly, spatial TEC in differential maps for suspected days confirm EQ induce TEC variations during UT = 10–14 h (Figs. 4 and 6). One can see clearly the enhancement in TEC for 2–4 h in bi-hourly temporal plots, and subsequently in spatial maps acquired over epicenter. In spatial TEC maps, no abrupt electron clouds occurred during other UT hours, only due to geomagnetic storm TEC drifted over equator to follow normal eastward direction (Fig. 7).

In case of the Turkey's M_w 7.1 EQ, a strong geomagnetic storm was recorded having Dst < -100nT and $K_p > 5$ (Fig. 2). This geomagnetic storm also causes TEC anomaly in the analysis of GPS stations, which operated around EQ epicenter (Fig. 5). Spatial TEC maps manifest no profound electron cloud around epicenter and it follow normal equatorial path, which may not be influenced by epicenter (Fig. 7). These analyses show that it can be possible to differentiate EQ ionospheric anomaly from the one caused by geomagnetic storm, however, it still need more observation using ionospheric measurements from GPS and other techniques. In other studies, the coupling between the EQ and ionosphere has been also reported by the observations of different ionospheric parameters. For example, the EQ induced positive holes during the process of squeezed rocks within seismogenic zones could variate air ionization and as a result increase the conductivity (Freund et al., 2007). Once the process is initiated, the more

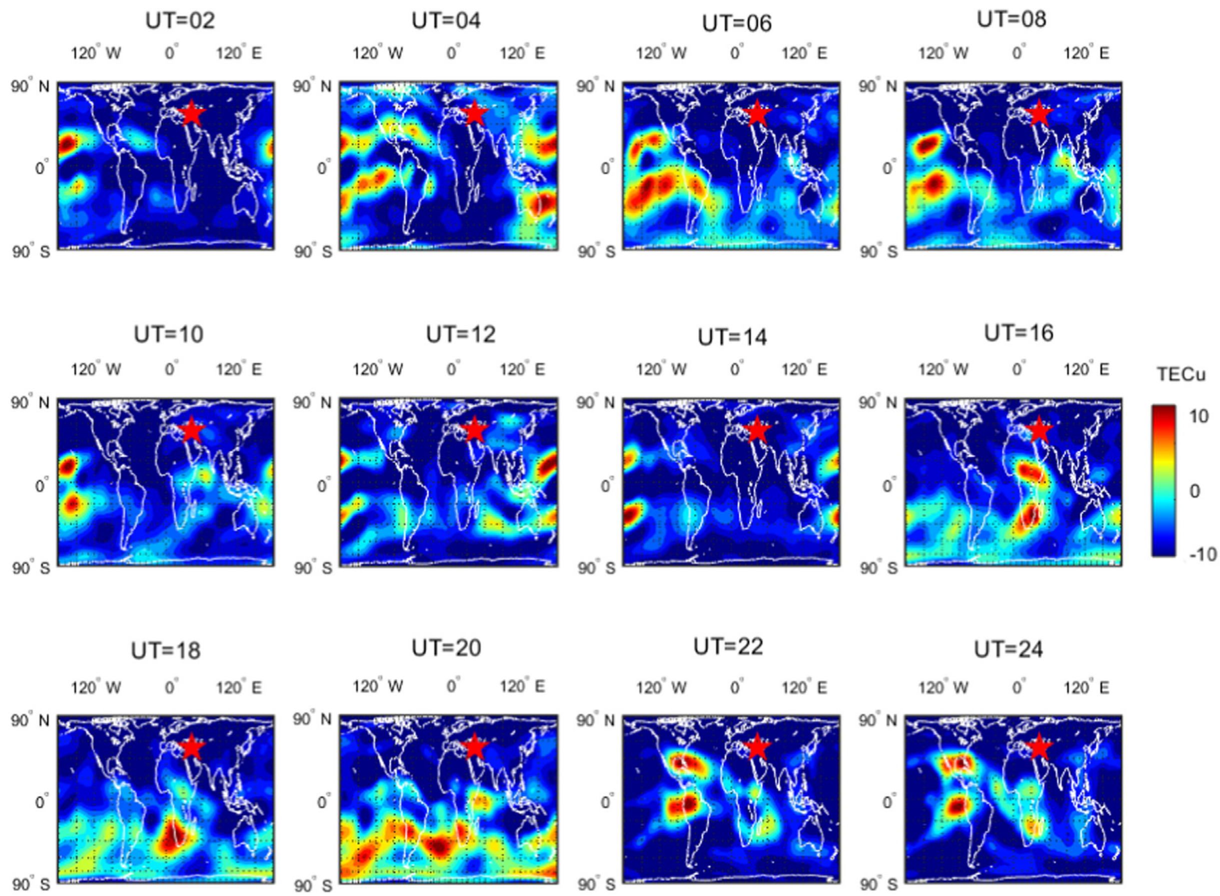

Fig. 7. Differential GIM TEC maps on anomalous day related to geomagnetic storm after Turkey EQ on October 25, 2011 (2 days after main shock).

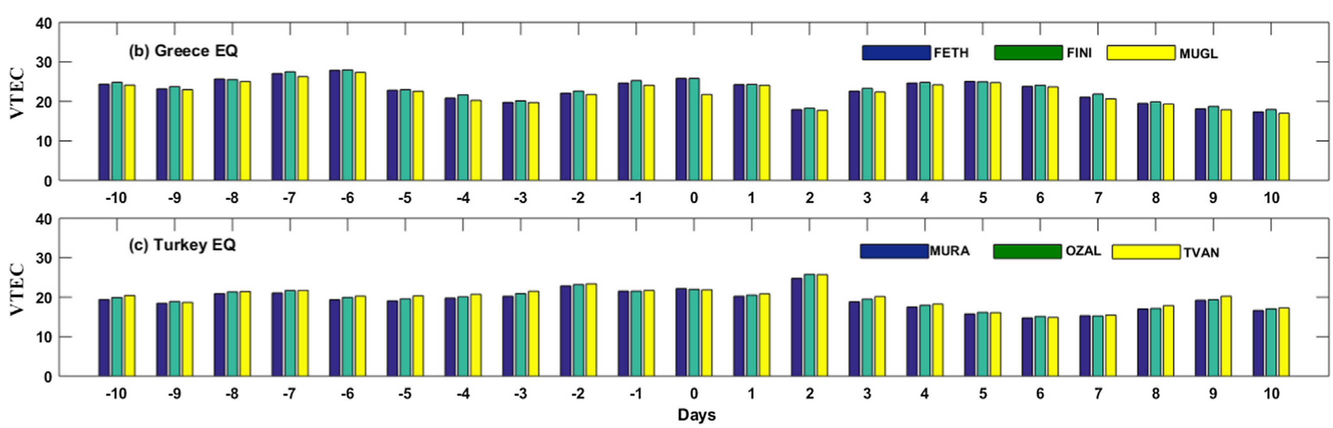

Fig 8. Comparison of daily VTEC for each EQ from three GPS ground-stations in Turkey within seismogenic zone for 10 days before/after.

influx of ions will further enhance the ionospheric potential at the lower end of ionosphere and then propagate upward.

Similarly, Pulinets and Ouzounov (2011) reported another mechanism of evolution of ionospheric anomalies by emission of Radon gas from EQ region and associated fault lineaments during seismic preparation period. They further explained that positive and negative ionospheric anomalies could occur due to increased and decreased air conductivity through Radon emanation over epicenter. It makes sense that positive anomalies in our study may be due to increased amount of gases from rupture zone during preparation period (see, e.g., Figs. 3 and 5). Ondoh (2003) reported formation of ion pairs and increase air conductivity by radioactive decay of Radon atoms via high energy alpha particles, which is ionized by initial emission from EQ prone region. Similarly, Rycroft et al. (2008) found seismo ionospheric anomalies as a result of modified air conductivity at Earth surface lead to abrupt changes in

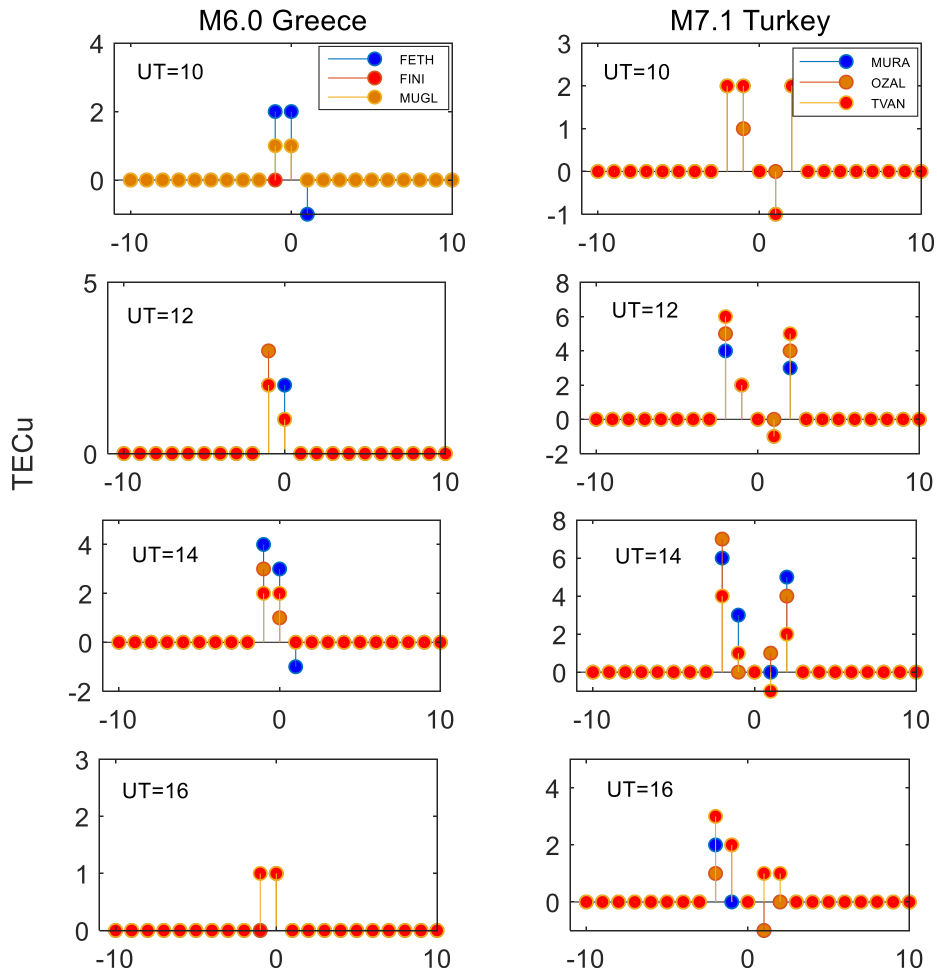

Fig. 9. Relative TEC during suspected UT = 10–16 h from three stations within EQ critical region and analysis of 10 before/after the main shock.

the ionosphere. Harrison et al. (2010, 2014) implemented this method to better explain the mechanism of generation and evaluation seismo ionospheric perturbations.

The coupling between lithosphere and ionosphere was demonstrated from numerical simulations by the use of atmospheric ionization and the resulting conductivity then can jointly trigger an upward electric current (Kuo et al., 2011). They further reported that the constant current flow from EQ epicenter can further stimulate the horizontal electric field at the bottom of ionosphere to initiate the vertical or zonal drift of ionospheric plasma following the magnetic field direction. Later, Kuo et al., (2014) introduced any arbitrary angle of potential difference to further trace the travel of electric current from epicenter towards ionosphere and it helped in improvement of lithosphere atmosphere ionosphere coupling system. Furthermore, they explained that increase/decrease in ionospheric perturbations can occur only above epicenter due to plasma flow direction. This definition of generation of positive anomalies in Kuo et al. (2014) completely agrees with the analyses in our study. We find significant positive anomalies in both spatial and temporal pattern around epicenter associated with $M_w \geq 6.0$ EQs. Besides temporal TEC, GIM TEC maps were used to detect abnormal EQ signal in previous studied to verify temporal TEC results (Shah and Jin, 2015; Tariq et al., 2019). However, there still needs more observations and analyses with durable satellite cluster for calculating different ionospheric indices.

5. Conclusion

In this paper, we have investigated seismo ionospheric anomalies with two $M_w \geq 6.0$ EQs detected by permanent GPS ground-stations in Turkey operating around epicenters. The findings of EQ induced TEC anomalies in both temporal and spatial analyses prove ionospheric variations for EQ forecasting. The main findings in this paper follow the previous finding of EQ related TEC anomalies within 5 days before main shock by Liu et al., (2004). In first case study of M_w 6.0, Greece EQ, significant ionosphere anomalies in temporal and spatial TEC is recorded on main shock day. Moreover, both VTEC and GIM TEC show significant anomalies 2 days before M_w 7.1 Turkey's EQ. We find evidences of temporal and spatial ionospheric anomalies within 5 days before main shock of the two EQs and these anomalies initiated during UT = 10 h on small scale, linger around epicenter on huge scale during UT = 12–14 h and lasted gradually until UT = 14 h. All these occur under quiet geomagnetic storm condition ($K_p < 3$ and $Dst = 0$ nT) during EQ preparation period. In the last case of the M_w 7.1 Turkey's EQ, a sudden geomagnetic storm occurred 2 days after the main shock day, it showed no association with epicenter and seismogenic zone during any UT hours and followed normal characteristically equatorial path. On the other hand, pre seismic anomalies in the form of dense electron clouds in spatial maps over epicenter occurred during UT = 10–14 h in case of Turkey's M_w 7.1 EQ. More analyses on TEC measurement with improved observations and dense cluster can effectively manifest more results.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgement

The authors are very grateful to USGS and ISGI for providing EQs and geomagnetic storm data. We are also thankful to International GNSS Service for TEC GIMs and permanent GPS ground-stations in Turkey for making available VTEC observations. We are also thankful to the Editor-in- Chief, Associate Editor and two anonymous reviewers for useful comments for furnishing the final manuscript.

References

Afraimovich, E.L., Astafyeva, E.I., 2008. TEC anomalies—Local TEC changes prior to earthquakes or TEC response to solar and geomagnetic activity changes?. Earth Planets Space 60, 961–966. https://doi.org/10.1186/BF03352851.

Ahmed, J., Shah, M., Zafar, W.A., Amin, M.A., Iqbal, T., 2018. Seismoionospheric anomalies associated with earthquakes from the analysis of the ionosonde data. J. Atmos. Sol.-Terr. Phys. 179, 450–458. https://doi.org/10.1016/j.jastp.2018.10.004.

Bartels, J., Heck, N.H., Johnston, H.F., 1939. The three-hour-range index measuring geomagnetic activity. J. Geophys. Res. 44 (4), 411. https://doi.org/10.1029/te044i004p00411.

Burešova, D., Laštovička, J., 2007. Pre-storm enhancements of foF2 above Europe. Adv. Space Res. 39, 1298–1303.

Ciraolo, L., Azpilicueta, F., Brunini, C., Meza, A., Radicella, S.M., 2007. Calibration errors on experimental slant total electron content (TEC) determined with GPS. J. Geod. 81, 111–120. https://doi.org/10.1007/s00190-006-0093-1.

Daneshvar, M.R.M., Freund, F.T., 2017. Remote sensing of atmospheric and ionospheric signals prior to the Mw 8.3 Illapel earthquake, Chile 2015. Pure Appl. Geophys. 174, 11–45. https://doi.org/10.1007/s00024-016-1366-0.

Dautermann, T., Calais, E., Haase, J., Garrison, J., 2007. Investigation of ionospheric electron content variations before earthquakes in southern California, 2003–2004. J. Geophys. Res. Solid Earth 112. https://doi.org/10.1029/2006JB004447.

Davis, T.N., Sugiura, M., 1966. Auroral electrojet activity index AE and its universal time variations. J. Geophys. Res. 71 (3), 785–801. https://doi.org/10.1029/jz071i003p00785.

Dobrovolsky, I., Zubkov, S., Miachkin, V., 1979. Estimation of the size of earthquake preparation zones. Pure Appl. Geophys. 117, 1025–1044. https://doi.org/10.1007/BF00876083.

Feltens, J., 2003. The international GPS service (IGS) ionosphere working group. Adv. Space Res. 31, 635–644.

Freund, F., Takeuchi, A., Lau, B.W.S., Al-Manaseer, A., Fu, C.C., Byrant, N.A., Ozounov, D., 2007. Stimulated infrared emission from rocks: assessing a stress indicator. Earth 2, 1–10.

Freund, F.T., Kulahci, I.G., Cyr, G., Ling, J., Winnick, M., Tregloan Reed, J., 2009. Air ionization at rock surfaces and pre-earthquake signals. J. Atmospheric Solar Terrestrial Phys. 71 (17–18), 1824–1834.

Fujiwara, H., Kamogawa, M., Ikeda, M., Liu, J.Y., Sakata, H., Chen, Y. I., Ofuruton, H., Muramatsu, S., Chuo, Y.J., Ohtsuki, Y.H., 2004. Atmospheric anomalies observed during earthquake occurrences. Geophys. Res. Lett. 31, L17110. https://doi.org/10.1029/2004GL019865.

Geller, R.J., Jackson, D.D., Kagan, Y.Y., Mulargia, F., 1997. Earthquakes cannot be predicted. Science 275 (5306), 1616.

Harrison, R.G., Aplin, K.L., Rycroft, M.J., 2010. Atmospheric electricity coupling between earthquake regions and the ionosphere. J. Atmos. Sol. Terr. Phys. 72, 376–381. https://doi.org/10.1016/j.jastp.2009.12.004.

Harrison, R.G., Aplin, K.L., Rycroft, M.J., 2014. Brief communication: Earthquake-cloud coupling through the global atmospheric electric circuit. Nat. Hazards Earth Syst. Sci. 14, 773–777. https://doi.org/10.5194/nhess-14-773-2014.

Heki, K., Enomoto, Y., 2013. Pre seismic ionospheric electron enhancements revisited. J. Geophys. Res. Space Phys. 118, 6618–6626. https://doi.org/10.1002/jgra.50578.

Hernández-Pajares, M., Juan, J., Sanz, J., Orus, R., Garcia-Rigo, A., Feltens, J., Komjathy, A., Schaer, S., Krankowski, A., 2009. The IGS VTEC maps: a reliable source of ionospheric information since 1998. J. Geod. 83, 263–275.

Ke, F., Wang, Y., Wang, X., Qian, H., Shi, C., 2016. Statistical analysis of seismo-ionospheric anomalies related to Ms> 5.0 earthquakes in China by GPS TEC. J. Seismolog. 20, 137–149. https://doi.org/10.1007/s10950-015-9516-x.

Klobuchar, J.A., 1987. Ionospheric time-delay algorithm for single-frequency GPS users. IEEE Trans. Aerosp. Electron. Syst. 23, 325–331. https://doi.org/10.1109/TAES.1987.310829.

Klotz, S., Johnson, N.L., 1983. Encyclopedia of Statistical Sciences. John Wiley and Sons.

Kon, S., Nishihashi, M., Hattori, K., 2011. Ionospheric anomalies possibly associated with $M \geqslant 6.0$ earthquakes in the Japan area during 1998–2010: Case studies and statistical study. J. Asian Earth Sci. 41, 410–420. https://doi.org/10.1016/j.jseaes.2010.10.005.

Kuo, C.L., Huba, J.D., Joyce, G., Lee, L.C., 2011. Ionosphere plasma bubbles and density variations induced by pre-earthquake rock currents and associated surface charges. J. Geophys. Res. 116, A10317. https://doi.org/10.1029/2011JA016628.

Kuo, C.L., Lee, L.C., Huba, J.D., 2014. An improved coupling model for the lithosphere-atmosphere-ionosphere system. J. Geophys. Res. Space Phys. 119, 3189–3205. https://doi.org/10.1002/2013JA019392.

Le, H., Liu, J.Y., Liu, L., 2011. A statistical analysis of ionospheric anomalies before 736 M6. 0+ earthquakes during 2002–2010. J. Geophys. Res. Space Phys. 116. https://doi.org/10.1029/2010JA015781.

Li, H., Yuan, Y., Li, Z., Huo, X., Yan, W., 2012. Ionospheric Electron concentration imaging using combination of LEO satellite data with ground-based GPS observations over China. IEEE Trans. Geosci. Remote Sens. 50 (5), 1728–1735. https://doi.org/10.1109/TGRS.2011.2168964.

Liu, J.Y., Chen, Y., Chuo, Y., Tsai, H., 2001. Variations of ionospheric total electron content during the Chi-Chi earthquake. Geophys. Res. Lett. 28, 1383–1386. https://doi.org/10.1029/2000GL012511.

Please cite this article as: M. Shah, S. Inyurt, M. Ehsan et al., Seismo ionospheric anomalies in Turkey associated with $M_w \geq 6.0$ earthquakes detected by GPS stations and GIM TEC, Advances in Space Research, https://doi.org/10.1016/j.asr.2020.03.005

Liu, J.Y., Chuo, Y.J., Shan, S.J., Tsai, Y.B., Chen, Y.I., Pulinets, S.A., 2004. Pre-earthquake ionospheric anomalies registered by continuous GPS TEC measurements. Ann. Geophys. 22, 1585–1593.

Liu, J.Y., Chen, Y.I., Chuo, Y.J., Chen, C.S., 2006. A statistical investigation of pre-earthquake ionospheric anomaly. J. Geophys. Res. Space Phys. 111, A05304. https://doi.org/10.1029/2005JA011333.

Liu, J., Hernandez-Pajares, M., Liang, X., An, J., Wang, Z., Chen, R., Hyyppä, J., 2016. Temporal and spatial variations of global ionospheric total electron content under various solar conditions. J. Geod. 91 (5), 485–502. https://doi.org/10.1007/s00190-016-0977-7.

Loewe, C.A., Prölss, G.W., 1997. Classification and mean behavior of magnetic storms. J. Geophys. Res. Space Phys. 102 (A7), 14209–14213. https://doi.org/10.1029/96ja04020.

Ondoh, T., 2003. Anomalous Sporadic-E layers observed before M7.2 Hyogo-ken Nanbu earthquake; Terrestrial gas emanation model. Adv. Polar Upper Atmos. Res. 17, 96–108.

Píša, D., Parrot, M., Santolík, O., 2011. Ionospheric density variations recorded before the 2010 Mw 8.8 earthquake in Chile. J. Geophys. Res.: Space Phys. 116. https://doi.org/10.1029/2011JA016611.

Pulinets, S., Legen'ka, A., 2003. Spatial–temporal characteristics of large scale disturbances of electron density observed in the ionospheric f-region before strong earthquakes. Cosm. Res. 41, 221–230. https://doi.org/10.1023/A:1024046814173.

Pulinets, S., Ouzounov, D., Ciraolo, L., Singh, R., Cervone, G., Leyva, A., Dunajecka, M., Karelin, A.V., Boyarchuk, K.A., Kotsarenko, A., 2006. Thermal, atmospheric and ionospheric anomalies around the time of the Colima M7.8 earthquake of 21 January 2003. Ann. Geophys. 24, 835–849. https://doi.org/10.5194/angeo-24-835-2006.

Pulinets, S., Ouzounov, D., 2011. Lithosphere-atmosphere-ionosphere coupling (LAIC) model – An unified concept for earthquake precursors validation. J. Asian Earth Sci. 41, 371–382. https://doi.org/10.1016/j.jseaes.2010.03.005.

Rishbeth, H., 2007. Do earthquake precursors really exist? Eos. Trans. American Geophys. Union 88 (29), 296.

Roma-Dollase, D., Hernández-Pajares, M., Krankowski, A., Kotulak, K., Ghoddousi-Fard, R., Yuan, Y., Li, Z., Zhang, H., Shi, C., Wang, C., 2018. Consistency of seven different GNSS global ionospheric mapping techniques during one solar cycle. J. Geod. 92, 691–706.

Rycroft, M.J., Harrison, R.G., Nicoll, K.A., Mareev, E.A., 2008. An overview of Earth's global electric circuit and atmospheric conductivity, Space Sci. Rev. 137, 83–105. https://doi.org/10.1007/s11214-008-9368-6.

Saroso, S., Liu, J.Y., Hattori, K., Chen, C.H., 2008. Ionospheric GPS TEC anomalies and M≥ 5.9 earthquakes in indonesia during 1993-2002, terrestrial. Atmospheric Oceanic Sci. 19.

Shah, M., Jin, S., 2015. Statistical characteristics of seismo-ionospheric GPS TEC disturbances prior to global Mw≥5.0 earthquakes (1998–2014). J. Geodyn. 92, 42–49.

Shah, M., Tariq, M.A., Ahmad, J., Naqvi, N.A., Jin, S., 2019a. Seismo ionospheric anomalies before the 2007 M7.7 Chile earthquake from GPS TEC and DEMETER. J. Geodyn. 127, 42–51. https://doi.org/10.1016/j.jog.2019.05.004.

Shah, M., Tariq, M.A., Naqvi, N.A., 2019b. Atmospheric anomalies associated with Mw>6.0 earthquakes in Pakistan and Iran during 2010-2017. J. Atmos. Sol.-Terr. Phys. 191. https://doi.org/10.1016/j.asr.2017.07.007 105056.

Tariq, M.A., Shah, M., Hernández-Pajares, M., Iqbal, T., 2019. Pre-earthquake ionospheric anomalies before three major earthquakes by GPS-TEC and GIM-TEC data during 2015–2017. Adv. Space Res. 63, 2088–2099. https://doi.org/10.1016/j.asr.2018.12.028.

Thomas, J., Huard, J., Masci, F., 2017. A statistical study of global ionospheric map total electron content changes prior to occurrences of M≥ 6.0 earthquakes during 2000–2014. J. Geophys. Res. Space Phys. 122, 2151–2161. https://doi.org/10.1002/2016JA023652.

Yao, Y., Chen, P., Zhang, S., Chen, J., Yan, F., Peng, W., 2012. Analysis of pre-earthquake ionospheric anomalies before the global M= 7.0+ earthquakes in 2010. Nat. Hazards Earth Syst. Sci. 12, 575–585. https://doi.org/10.5194/nhess-12-575-2012.

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